



Quantum-resilient blockchain-enabled secure communication framework for connected autonomous vehicles using post-quantum cryptography

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ABSTRACT

Connected and Autonomous Vehicles (CAVs) are pivotal to the evolution of Intelligent Transportation Systems (ITS), offering enhanced connectivity and automation. However, the emergence of quantum computing poses significant security challenges to existing cryptographic protocols. This study addresses these challenges by proposing a hybrid security approach that combines Kyber Post-Quantum Cryptography (PQC) with an Adaptive Grouping Score-based Practical Byzantine Fault Tolerance (AGS-PBFT) blockchain mechanism. The key contribution of this study is the integration of lattice-based Kyber PQC, which is resistant to quantum attacks, with a dynamically adaptive AGS-PBFT blockchain. This integration aims to secure vehicular communications by ensuring data integrity, authenticity, and confidentiality while enhancing the scalability and efficiency of consensus processes in dynamic CAVs environments.

The proposed hybrid approach has been validated through extensive simulations using the OMNET++ and SUMO simulators. The simulation results demonstrate that our approach outperforms existing methods, achieving lower latency, reduced computation costs, higher throughput, and enhanced security levels. Overall, the findings and methodologies presented in the paper can serve as a valuable reference for researchers and practitioners aiming to enhance the security and efficiency of CAVs in the era of quantum computing.

1. Introduction

Recent advancements in the Internet of Things (IoT) and Industry 4.0 have significantly transformed various sectors [1], [2], [3], where Connected and Autonomous Vehicles (CAVs) have emerged as an essential component of Intelligent Transportation Systems (ITS) [4]. These developments have laid the foundation for transformative automotive technologies that integrate enhanced connectivity and automation. The autonomy in CAVs is powered by advanced onboard units that exhibit sensing, communication, and network capabilities [5]. These units enable interactions with vehicles, pedestrians, roadside units (RSUs), and network infrastructure through wireless communication and sensing technologies according to established protocols like IEEE 802.11p and cellular technologies [6], [7]. This connectivity facilitates various transportation services, including autonomous driving, traffic congestion warnings, collision avoidance, emergency responses, infotainment, route navigation, and intelligent parking solutions to improve traffic efficiency and road safety [8], [9].

Although Connected and Autonomous Vehicles (CAVs) provide substantial benefits, but their widespread adoption faces numerous chal-

lenges. One of the primary concerns is the susceptibility of vehicular communications in wireless environments. Fig. 1 illustrates a range of communication links in an existing edge-based CAVs environment. These intelligent CAVs are equipped with embedded processors, advanced connectivity, and automated technologies, integrating various sensors, Electronic Control Units (ECUs), cameras, and Onboard Units (OBUs). They facilitate interaction with each other through specific communication links in a Vehicle-to-Everything (V2X) environment. These sensors gather substantial data from the surroundings, which is rapidly processed by RSUs and distributed edge servers to support effective decision-making. Despite these technological advancements, the connectivity of CAVs to the Internet using V2X links poses significant security vulnerabilities. As depicted in Fig. 1, the existing architecture relies on a centralized model where a central server coordinates all the interconnected CAVs and RSUs, maintaining bi-directional communication with the cloud [10]. This centralized architecture introduces a single point of failure, making the system vulnerable to several security threats. Furthermore, data transmitted over these links can be intercepted and tampered with by adversaries. Malicious entities within the public communication channels may impersonate legitimate nodes,

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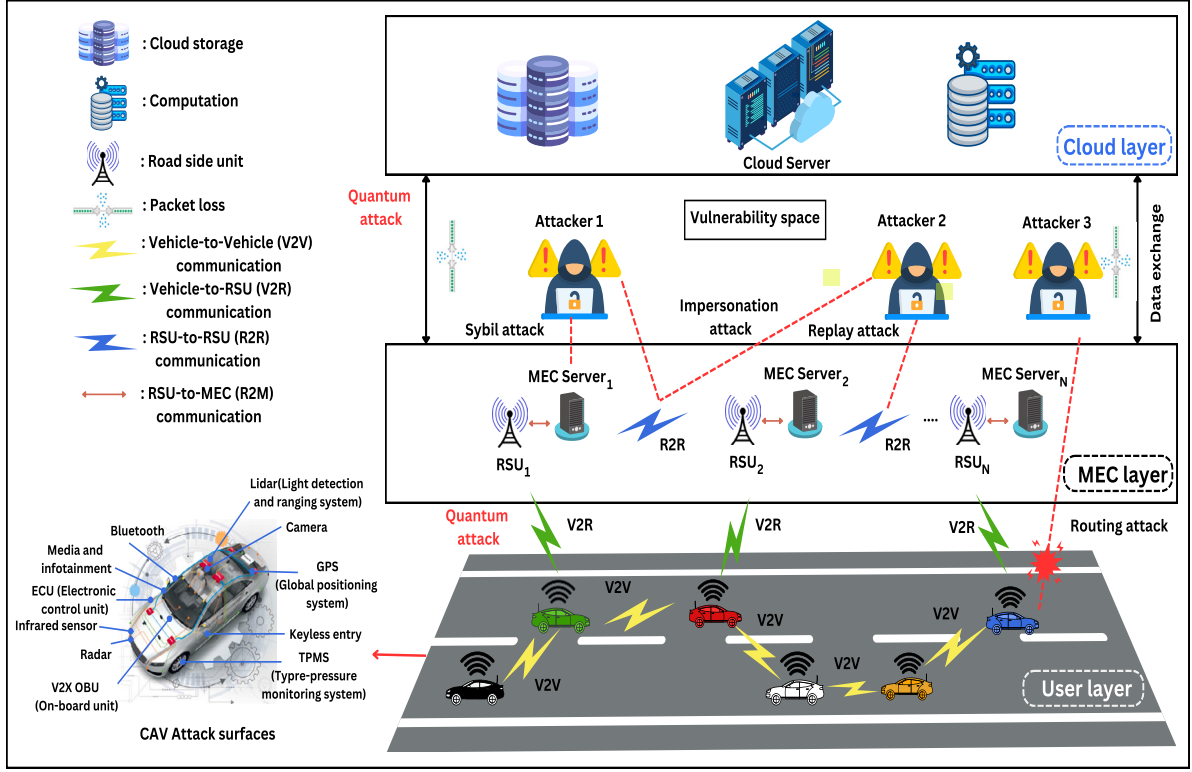


Fig. 1. Attack vectors in existing edge-based connected and autonomous vehicle environment.

initiating data tampering, eavesdropping, man-in-the-middle attacks, replay attacks, impersonation, and Denial-of-Service (DoS) attacks. These attacks can disrupt the network's standard functionality and compromise the integrity of the transmitted data.

To ensure the security of shared information, standards such as American IEEE 1609.2 [11] and European Telecommunications Standards Institute (ETSI) [12] have been proposed along with robust encryption protocols to improve authentication and message verification [13]. Public-key Cryptography (PKC) provides the framework for secure communication between automotive devices, utilizing cryptosystems like Rivest–Shamir–Adleman (RSA) and the Elliptic Curve Digital Signature Algorithm (ECDSA). These cryptosystems are based on solving complex mathematical problems, such as large integer factorization and discrete logarithms. However, the impending development of quantum computing poses significant threats to these existing cryptosystems. As shown in Table 1, quantum algorithms like those developed by Shor [14] and Grover [15] can breach existing cryptosystems such as RSA, ECDSA, and SHA hash functions in the near future. This imminent vulnerability, combined with the rapid advancement of quantum computing and the widespread deployment of CAVs, necessitates the development of efficient Quantum-proof authentication protocols based on mathematically challenging problems to provide resistance against quantum attacks.

In 2016, the National Institute of Standards and Technology (NIST) initiated a program to develop Post-Quantum Cryptography (PQC) primitives to defend against quantum attacks [16]. Reliable cryptographic solutions are proposed using NP-hard lattice problems, which employ lattice structures consisting of repeating points across multi-dimensional spaces. The key NP-hard problems in these cryptosystems include finding the Shortest non-zero Vector (SVP), solving the Short Integer Solutions (SIS), and the Closest Vector Problems (CVP), which have additional algebraic structure to provide quantum resistance [17]. Moreover, Learning With Errors (LWE) and its variations, such as Ring-LWE and LP-LWE, form the basis of these lattice-based cryptosystems [18]. However, amongst them, CRYSTAL-Kyber, based on the Module-LWE (MLWE) problem [19], has emerged as one of the most promising public key

encryption and key encapsulation mechanisms (PKE/KEM). Its optimal balance between key size, security, and computational efficiency can provide substantial resistance in CAVs to ensure secure communication and resistance against quantum attacks [20], [21].

Although the security of CAVs can be preserved using quantum-resistant protocols, the distributed secure storage of extensive CAVs data remains vulnerable. Integrating blockchain technology [22] can substantially reinforce the security framework for CAVs systems by addressing centralized vulnerabilities and eliminating single points of failures that quantum-enabled adversaries might exploit [23]. Blockchain provides a decentralized, immutable, secure, and anonymous approach to data management, which ensures transparency, auditability, and tamper-resistance of data transactions, vital in maintaining the integrity of vehicle communications and operations [24]. Therefore, integrating lattice-based cryptography with blockchain technology secures communication channels against quantum attacks and establishes a robust infrastructure for the efficient management of CAVs. Thus, in this study, we proposed a Kyber-based post-quantum cryptography approach that integrates with the Practical Byzantine Fault Tolerance (PBFT) blockchain with an adaptive grouping score mechanism for CAVs to enable secure communication, enhancing trust and reliability in future ITS deployments where security and data integrity are paramount.

1.1. Motivation of this study

The communication of traffic information between CAVs relies on wireless technology, making it susceptible to potential attacks, thus compromising the integrity of information essential for safe driving. These vulnerabilities highlight the need for robust authenticated key exchange (AKE) and key encapsulation mechanism (KEM) frameworks incorporating post-quantum cryptographic solutions to ensure secure communication and tamper-proof storage. Despite advancements in authenticated key agreement schemes, several limitations and inadequacies remain. However, the rise of quantum computing introduces unprecedented challenges as conventional cryptographic methods, upon

Table 1

Summary of existing cryptographic algorithms and impact by quantum computing.

Symmetric cryptography			
Algorithm	Cryptographic Function	Security Level (bits)	Quantum impact
AES-128	Symmetric encryption	128	Grover
AES-256	Symmetric encryption	256	Grover
SALSA-20	Stream Cipher	256	Grover
SHA-256	Hash Function	256	Grover
SHA-3	Hash Function	256	Grover
Asymmetric cryptography/ Public-key cryptography			
RSA-3072	Encryption/Signature	128	Shor (Broken)
DH-3072	Key exchange	128	Shor (Broken)
ECDSA-256	Signature	128	Shor (Broken)
ECDH-256	Key exchange	128	Shor (Broken)

which current CAVs communication protocols heavily rely, are vulnerable to quantum algorithms. To address this proactively, an emphasis on **Lattice-based Cryptography (LBC)** [17], particularly the Kyber [25], is gaining importance due to its robustness against quantum attacks. LBC is built on complex mathematical problems such as the Learning With Errors (LWE) and its variant Ring-Learning With Errors (Ring-LWE), which offers security against existing and quantum computers [26]. Kyber, based on the Module-LWE (MLWE) problem, is versatile in handling key exchanges, digital signatures, and homomorphic encryption, making it ideal for long-term security in critical infrastructures and safety-critical environments like CAVs. As quantum computing advances, the imperative for implementing post-quantum cryptographic solutions becomes increasingly urgent to secure the future of CAVs and ensure unbreakable security links within vehicular networks.

Furthermore, integrating these cryptographic solutions with a **decentralized Practical Byzantine Fault Tolerance (PBFT) blockchain** based on an adaptive score grouping mechanism enhances the overall security framework, allowing for secure management of communications and transactions within the CAVs environment. This approach secures CAVs from current cybersecurity threats and prepares them for future quantum challenges, ensuring enhanced safety, privacy, and operational efficiency in increasingly smart vehicular environments.

1.2. Why AGS-PBFT blockchain-based post-quantum cryptography (PQC) is needed?

The Adaptive Grouping Score-based Practical Byzantine Fault Tolerance (AGS-PBFT) blockchain-enabled approach, integrated with Kyber PQC, is highly effective for ensuring data integrity and information storage and transmission transparency without relying on a central entity in Connected and Autonomous Vehicles (CAVs). Due to limited computing resources and potential unfairness, existing consensus algorithms like Proof of Work (PoW), Proof-of-Stake (PoS), and Delegated Proof-of-Stake (DPoS) are not suitable for the current CAVs environment. Practical Byzantine Fault Tolerance (PBFT), although the most potential consensus protocol, is based on a specific consensus mechanism [27], where every node on the chain participates in the consensus process but faces challenges in dynamic CAVs environments due to high communication overhead and consensus delays. AGS-PBFT addresses these issues by employing a grouping score mechanism that dynamically updates consensus nodes, thereby improving reliability and scalability.

The adaptive grouping score mechanism in AGS-PBFT works by initially assigning scores to nodes and grouping them into consensus and candidate sets. The consensus nodes participate actively, while the candidate nodes update their local states without participating. The nodes are re-evaluated periodically and reassigned based on their scores, with high-performing nodes assigned to the consensus set and low-performing ones placed in the candidate set. This dynamic adjustment helps maintain an efficient and reliable consensus process, even with

changing network conditions. By using this adaptive grouping score mechanism, AGS-PBFT optimizes resource utilization, reduces communication overhead, and enhances the performance and security of the blockchain network in CAVs. Additionally, incorporating Kyber PQC, a lattice-based cryptographic approach that leverages Kyber's IND-CCA2 [25] security, provides strong resistance against quantum adversaries. Thus enhancing the overall security of CAVs communications against future quantum threats. This integration reduces consensus latency, mitigates block congestion, and increases transaction throughput while maintaining robust security in CAVs.

1.3. Key contributions

This study proposed an integrated post-quantum cryptography and blockchain approach designed to enhance security in Connected and Autonomous Vehicles (CAVs) by providing resistance against quantum threats. The key contributions of this work are summarized as follows:

- We proposed a **Kyber-PQC-based-AGS-PBFT approach** designed to address the critical issue of **data security** in CAVs by leveraging Kyber, a lattice-based post-quantum cryptography (PQC) combined with an Adaptive Grouping Score-based Practical Byzantine Fault Tolerance (AGS-PBFT) blockchain consensus mechanism. This **framework** ensures resistance against both quantum and existing threats.
- The **AGS-PBFT mechanism** dynamically adjusts the grouping of nodes based on performance metrics. This approach enhances the scalability and efficiency of the consensus process while also **improving fault tolerance** in dynamic CAVs environments.
- We validated the effectiveness of the proposed approach by comparing it with Quantum Defended Blockchain Assisted Conditional Privacy Preserving Data Authentication (QBCPDA) [28], Lattice-based Two-Party Authenticated Key Agreement (LB-ID-2PAKA) [29], Mutual Authentication and Key Management (MAKMS-IoV) [30], and Semi-Trusted Authority-based Authentication (STAA-VANET) [31]. Our results demonstrate significant improvements, offering **high security and resilience** against existing and future quantum attacks while maintaining low computation costs, latency, and **high throughput**. These enhancements improve the overall efficiency and reliability of CAVs communications.

1.4. Article structure

The remainder of the paper is structured as follows: Section 2 discusses the related work. Section 3 discusses some preliminaries related to lattice PQC, followed by the system framework of Kyber-PQC-based-AGS-PBFT described in Section 4. Section 5 describes the secure communication mechanism, and Section 6 elaborates on the proposed approach's security analysis. The performance evaluation is discussed in

Section 7, and finally, the conclusion with future direction is presented in Section 8.

2. Related work

We review the existing and quantum-proof access control, key management, and authentication frameworks developed for Intelligent Transportation Systems (ITS) in vehicular communications. Furthermore, we assess their effectiveness in the context of the most recent advances in communication and security technologies [32].

Information security faces significant challenges as Connected and Autonomous Vehicles (CAVs) evolve. The author in [33] conducted a comprehensive survey in the area of CAVs security and identified a wide range of vulnerabilities in CAVs' network connectivity, including attacks by other users and access control issues (like illegal copying and unauthorized access), availability issues (like replay and relay attacks), and privacy issues (like impersonations and repudiations). The authors in [34] proposed a reputation-value-focused blockchain-based system for trust management in the Internet of Vehicles (IoV). In another study, the authors introduced Blockchain, a blockchain-based message revocation and accountability framework to ensure safe communication in IoV [35]. However, these systems do not address the reliability of data transmission, which would be compromised by in-vehicle ECU attacks. Furthermore, CreditCoin, a privacy-preserving blockchain-based system for managing reputation management and reward announcement systems for smart vehicles, was proposed by authors in [36]. Their proposal faces significant challenges related to storage and scalability and is vulnerable to collusion attacks due to its reliance on threshold authentication. Furthermore, several developments have been made in certificate-based batch verification techniques [37], [38], data sharing [39], privacy protection [40], [41], and authentication mechanisms to address the real identities, privacy, and data injection attacks in vehicles. However, due to the time-consuming procedures involved in the rapid communication of information between vehicles and RSU, the substantial storage demands for cryptographic keys, and the requirement for massive certificates, most of these mechanisms are not appropriate for the broader CAVs environment.

To mitigate the burden of certificate management, the authors in [16] developed a certificateless signature scheme for vehicular networks to guarantee identity privacy and satisfy network constraints by utilizing semi-trusted authority and the RSUs for identity management. It incorporates a self-healing key distribution technique with a certificateless signature that restricts group key generation to authorized vehicles. The authors in [23] developed a mutual authentication protocol for the Internet of Vehicles (IoV) to address security challenges by preserving secure communication between vehicles and infrastructure. Their proposed protocols rely on a trusted third party and a combination of symmetric and asymmetric encryption techniques. However, despite their robustness, these classical public-key-based solutions remain vulnerable to quantum attacks, particularly from quantum computers using Shor's algorithm. Thus, the existing authentication approaches for CAVs include lightweight protocols [42], which utilize simple cryptographic primitives such as XOR, hash functions, and symmetric encryption due to their minimum execution time. While efficient, these lightweight approaches are vulnerable to sophisticated attacks. To further enhance security, more robust public-key-based solutions employing cryptosystems such as RSA [43] and Elliptic Curve Cryptography (ECC) [44], [45] have been developed. Many of these existing approaches have either used Diffie-Hellman (DH) [46] or prime factorization-type hard problems, which are vulnerable to futuristic technologies like quantum computers, potentially compromising these established cryptosystems.

Thus, Quantum cryptography must be adopted in the approaching era of quantum computing to provide resilience against existing and future quantum attacks. Lattice-based cryptography has recently been proposed as an alternate approach to provide resistance against quantum attacks [47], [48]. The authors in [29] proposed a lattice-based, two-

party authenticated key agreement protocol using identity-based cryptography (IBC). The approach leverages lattice hard problems, which could resist quantum computing and IBC could eliminate the requirement of certificate management. [28] introduces a quantum-resistant blockchain-assisted conditional privacy-preserving data authentication technique for the IoVs to ensure secure batch verification. This technique utilizes lattice cryptography to safeguard against data forgery and quantum attacks.

In summary, Table 2 compares the security features and attack resilience of the proposed approach with other classical and lattice-based approaches. As shown in Table 2, classical approaches offer robust security features but lack resistance to quantum attacks. In contrast, lattice-based solutions, which are still evolving, offer quantum-proof capabilities. Our proposed approach is based on lattice-based cryptography, ensuring resilience against existing and future quantum attacks in CAVs.

2.1. Comparison with state-of-the-art approaches

Despite the significant progress in securing CAVs against cyber threats, the existing access control, key management, and authentication frameworks exhibit critical limitations, particularly with the advent of quantum computing. Existing studies, including [30], [31], and [44], rely on classical cryptographic methods such as ECC, RSA, and DH, which are vulnerable to quantum attacks as a result of Shor's algorithm. Our proposed approach employs Kyber-based PQC, leveraging lattice-based cryptography to ensure resistance against classical and quantum adversaries, offering future-proof security. Furthermore, existing blockchain-assisted CAVs systems utilize static consensus mechanisms such as PoW, PoS, and PBFT, necessitating the participation of all nodes in the consensus, resulting in high computational overhead, increased latency, and limited scalability. To address this, our proposed AGS-PBFT mechanism dynamically reassigns nodes into consensus and candidate groups based on real-time performance scores, reducing computational overhead, lowering block latency, and improving throughput. Moreover, while most of the prior approaches in CAVs focus on mitigating known threats like impersonation, replay attacks, and eavesdropping, they fail to account for emerging threats posed by quantum adversaries. Our dual-layer protection mechanism combines Kyber PQC for quantum resistance with the AGS-PBFT framework for adaptive and secure consensus, ensuring resilience against classical, quantum, and consensus-based threats. Thus, our proposed approach provides enhanced resistance against evolving attacks while improving fault tolerance and scalability within large, dynamic CAV networks, as analyzed in Section 6.

3. Preliminaries

Lattice-based cryptography is a variant of cryptography that leverages mathematical structures, which are geometric structures composed of infinite grid points in multi-dimensional space, to design cryptographic primitives and protocols. These cryptographic systems are based on complex lattice problems, which offer several benefits, including robust security and resistance to quantum attacks. Many lattice-based cryptographic schemes rely on the hardness of mathematical problems such as Learning With Errors (LWE) or Nth Degree Truncated Polynomial Ring Units (NTRU). Notably, a specific variant of these lattice problems called Module-LWE (MLWE) is used by cryptography systems such as Kyber [19].

3.1. The module learning with errors (MLWE)

The Module Learning with Errors (MLWE) is a variant of the Learning With Errors (LWE) problem and is related to the RLWE problem [49]. Assume that the ring of polynomials over integers n and q is

Table 2

Comparison of security features and attack resilience in existing and proposed approach.

Features/Attack resilience	[28]	[29]	[30]	[31]	[44]	[45]	[46]	Ours
Anonymity	✓	✓	✓	✓	×	×	×	✓
Unlinkability	✓	✓	✓	✓	×	×	×	✓
Simple public key revocation	✓	×	✓	×	×	×	×	✓
Perfect forward secrecy	✓	✓	✓	✓	×	×	×	✓
No key escrow	✓	✓	✓	✓	×	×	×	✓
Quantum-proof	✓	✓	×	×	×	×	×	✓
Resistance to MITM Attack	✓	✓	✓	✓	✓	✓	✓	✓
Authentication	✓	✓	✓	✓	✓	✓	✓	✓
Data Integrity	✓	✓	✓	✓	✓	✓	✓	✓
Confidentiality	✓	✓	✓	✓	✓	✓	✓	✓
Non-repudiation	✓	✓	✓	✓	✓	✓	✓	✓
Replay attack	✓	✓	✓	✓	✓	✓	✓	✓
Impersonation	✓	✓	✓	✓	✓	✓	✓	✓
Diffie-Hellman (DH) Problem	✓	✓	✓	✓	✓	✓	✓	✓
Tamper Resistance	✓	×	×	×	×	×	×	✓
Conditional Traceability	✓	✓	✓	✓	×	×	×	✓
Dynamic Group Management	×	×	×	×	×	×	×	✓
Elliptic Curve Cryptography (ECC)	×	×	✓	✓	✓	✓	✓	×
Post Quantum Cryptography (NTRU)	✓	✓	×	×	×	×	×	✓
Lattice-based Cryptography	✓	✓	×	×	×	×	×	✓

$R_q = \mathbb{Z}_q[X]/(X^n + 1)$, with $X^n + 1$ denoting the 2^n -th cyclotomic polynomial. To represent a $i \times j$ matrix within R_q , let $R_q^{i \times j}$ signify the matrix dimensions i, j , and m any positive integer. The difference between a uniform sample pair and a LWE sample $(A, B = A \cdot s + e)$ determines the LWE problem's hardness. In this case, the uniform distribution is represented by the public matrix $A \in R_q^{i \times j}$, the secret $s \in R_q^{j \times m}$, which is sampled from the binomial or Gaussian distribution, and the error $e \in R_q^{i \times m}$ is sampled from the same distribution. Furthermore, recovering the secret s from the LWE instance B , where A is known, is recognized to be a hard problem. Moreover, when the polynomial length n is constrained to a power-of-two value, the module structure of MLWE yields higher flexibility compared to RLWE. Consequently, PKE/KEM or signature schemes can be efficiently designed using LWE instances [18].

3.2. Symmetric primitives and number-theoretic transform in kyber

Kyber employs symmetric primitives, including a pseudorandom function PRF and extendable output functions (XOFs), specifically designed for efficient operation in the cryptographic protocol. KYBER utilizes two hash functions $H : \mathbb{B}^* \rightarrow \mathbb{B}^{32}$ and $G : \mathbb{B}^* \rightarrow \mathbb{B}^{32} \times \mathbb{B}^{32}$ along with a key-derivation function $KDF : \mathbb{B}^* \rightarrow \mathbb{B}^*$. These components are essential for the security and operations of the scheme, facilitating operations such as key derivation and data authentication [25].

Kyber, a cryptographic approach is based on the MLWE problem, operates on polynomials in the ring $\mathbb{Z}_q[X]/(X^{256} + 1)$. The choice of modulus $q = 3329$ is optimal for using Number Theoretic Transform (NTT), allowing fast polynomial multiplications which are critical in lattice-based cryptography. This choice facilitates rapid polynomial multiplications, effectively addressing one of the primary bottlenecks in LWE-based systems. Due to its performance and well-elaborated parameter choices, Kyber is one of the most promising PKE/KEM schemes.

The key operations used in the Kyber cryptosystem are as follows:

1. Key Generation: In the key generation process, a random seed d is generated and two parameters ρ and σ are derived using a pseudorandom function $G(\cdot)$ such that $(\rho, \sigma) \leftarrow G(d)$. Then, a public matrix $\hat{A}$ is generated using an extendable output function (XOF), defined as, $\hat{A} = \text{Parse}(XOF(\rho, i))$, $i \in \{0, \dots, k-1\}$. Following this, the secret vectors s and error term e are sampled from the centered binomial distribution B_η . Further, the public key pk is computed as, $pk = \hat{A} \cdot s + e$, where the secret key sk is stored as a tuple (s, e, pk, σ) .

2. Encryption and Decapsulation: In the encryption process, the sender uses the recipient's public key to generate a ciphertext c . To ensure security, noise e and randomness from the centered binomial distribution

B_n are used to prevent an adversary from recovering the message. The encapsulated ciphertext c is transmitted to the recipient.

During the decapsulation process, the recipient utilizes the secret key sk to recover the original message. Thus, this dual-process structure ensures confidentiality, integrity, and post-quantum resilience in the communication system.

3.3. Sampling from a binomial distribution

Noise in Kyber is sampled from a centred binomial distribution B_n , which has been defined for $n = 2$ or $n = 3$. This sampling process involves generating two sets of bits $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$, each bit being independently chosen, and computing their difference sum, $\sum_{i=1}^n (a_i - b_i)$.

This method ensures that each coefficient in the polynomials used within Kyber is sampled from B_n , providing the randomness and noise properties required for security in the MLWE problem.

4. Proposed system framework and threat model

This section outlines the system framework of our proposed framework and describes the threat model designed to meet the necessary security measures.

4.1. System framework

In this section, we present a quantum-resilient system framework designed to facilitate a structured and secure framework for managing interactions and data exchanges among various entities within the CAVs environment, as illustrated in Fig. 2. In the foreseeable future, Connected and Autonomous Vehicles (CAVs) are expected to operate with minimum or no human intervention. These vehicles are equipped with advanced connectivity and automated technologies, incorporating various sensors, Electronic Control Units (ECUs), cameras, and on-board units (OBUs) to enhance driving safety and passenger comfort. These sensors gather substantial data from the surroundings that are rapidly processed by distributed edge servers for effective decision-making. Despite these advancements, the connectivity of CAVs to the Internet poses significant security threats. Malicious entities may impersonate legitimate nodes, initiating attacks during data transit between edge servers, disrupting the standard functionality of the network, and compromising integrity.

To address these security challenges, we have highlighted the potential vulnerabilities and corresponding solutions within two architectural framework scenarios. These scenarios demonstrate the flow of

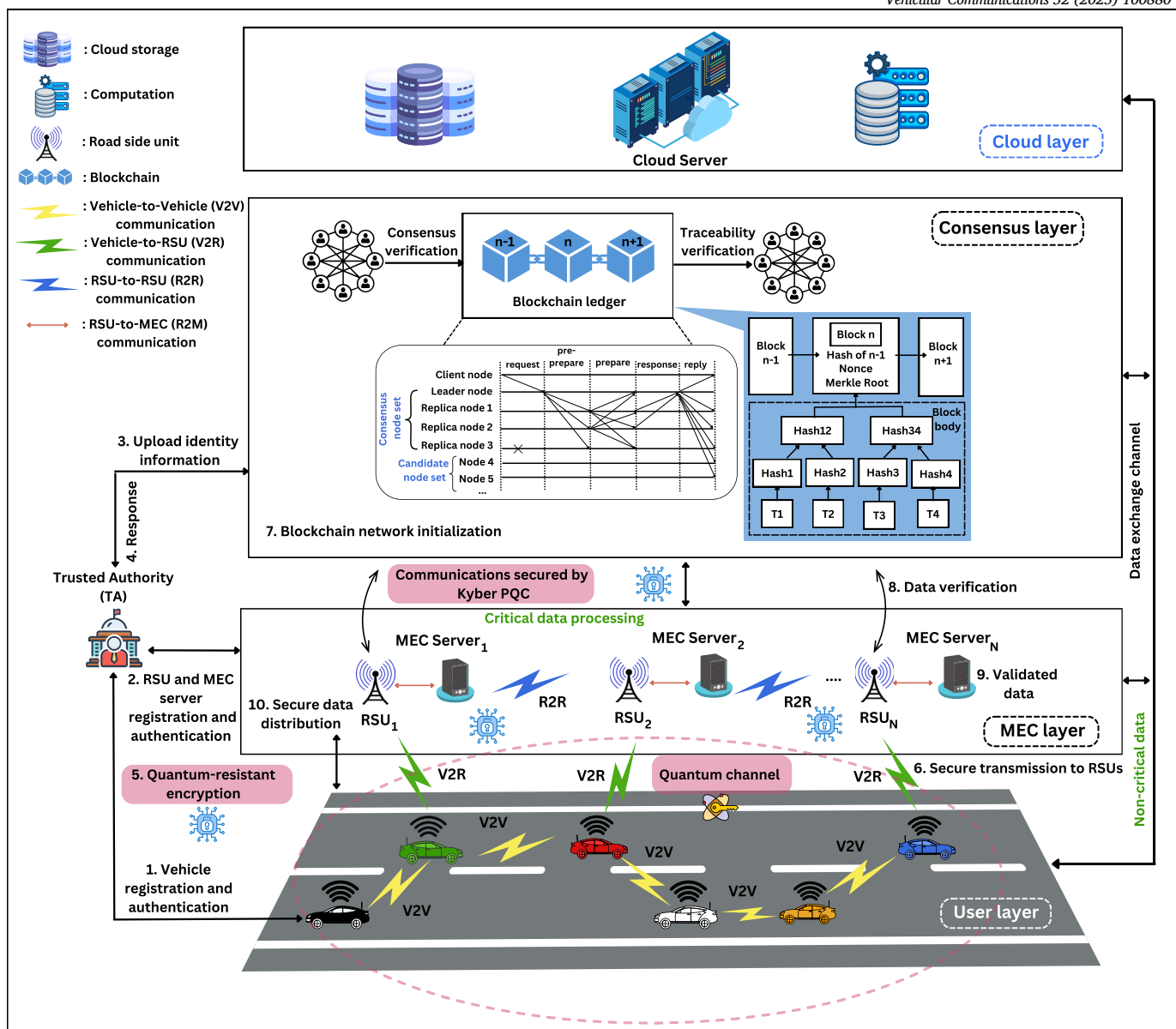


Fig. 2. Proposed system framework for quantum-resilient security in CAVs environment using Kyber-based post-quantum cryptography and AGS-PBFT blockchain.

data exchange between edge servers, as depicted in Fig. 1 and Fig. 2. Further, Fig. 1 illustrates an existing CAV system where communications between edge servers lack adequate security, safety, and reliability measures, making them susceptible to attacks that could compromise vehicle components or the entire system. We propose an enhanced framework integrating Kyber post-quantum cryptography (PQC) and blockchain-based systems to counteract these vulnerabilities. This integration aims to significantly improve communications security between nodes (such as CAVs and RSUs) by leveraging the inherent robust security capabilities of post-quantum cryptography and blockchain technology to protect against potential cyber and quantum threats effectively. The proposed blockchain-based Kyber PQC system framework encompasses several key components: Trusted Authority (TA), Connected and Autonomous Vehicles (CAVs) ($CAV_s, i = 1, 2, 3, \dots, n_{CAVs}$), Road-side Units ($RSU, i = 1, 2, 3, \dots, n_{RSU}$), Mobile Edge Computing (MEC) Servers ($MEC_i, i = 1, 2, 3, \dots, n_{MEC}$), and a cloud server. It is structured into various layers: user, MEC, consensus, and cloud. The flow diagram of the proposed system framework is illustrated in Fig. 3, and a detailed description of each layer, as depicted in Fig. 2, is as follows:

1. User layer: This layer encompasses Connected and Autonomous Vehicles (CAVs) equipped with Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) communication technologies, enabling dynamic data exchange and cooperative maneuvering. These communications are secured with quantum-resistant encryption protocols utilizing Kyber Crystal post-quantum cryptography (PQC). PQC ensures data integrity and confidentiality of critical information, such as speed, location, and navigation paths within the CAVs environment. These vehicles are equipped with sensors, cameras, and onboard units (OBUs) to facilitate real-time data collection for immediate decision-making, such as emergency braking, lane changes, and speed adjustments. These are crucial to identifying optimal routes, reducing traffic congestion, and preventing potential crashes.

2. MEC layer: This layer comprises Roadside units (RSUs) and Mobile Edge Computing (MEC) servers to facilitate real-time data processing directly at the network's edge, thus minimizing latency and enhancing decision-making for traffic management and emergency responses. MEC servers are strategically positioned near CAVs and RSUs, to handle and analyze vast volumes of data from CAVs to make rapid decisions. As shown in Fig. 1, attackers can potentially exploit vulnerabilities in

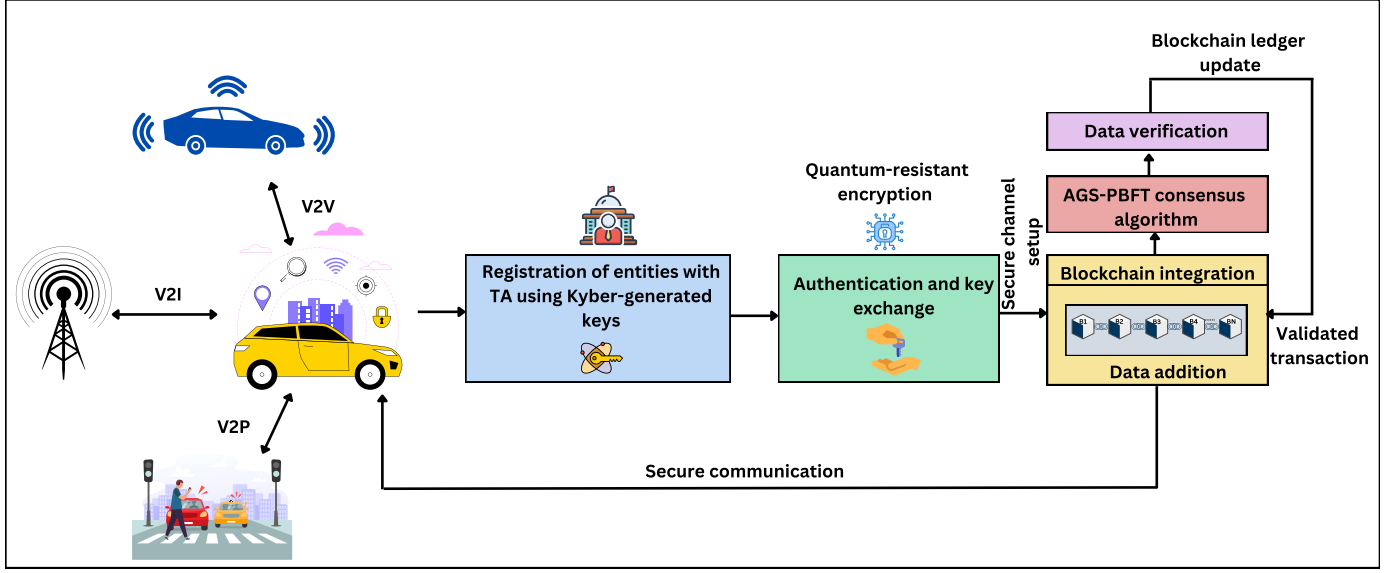


Fig. 3. Flow diagram of the proposed system framework for secure communication in CAVs.

the ports of the edge servers and intercept open wireless communication channels. Thus, each MEC server and RSU in the proposed system framework is integrated with the Kyber PQC to counteract these security threats, ensuring robust security against potential quantum threats and maintaining data integrity. Practical Byzantine Fault Tolerance (PBFT) based on an adaptive grouping score mechanism is employed to ensure that all transactions and data exchanges within the CAV network are consistent and tamper-proof, thereby enhancing the trustworthiness and reliability of the system.

3. Consensus layer: This layer in the CAVs framework employs Practical Byzantine Fault Tolerance (PBFT) based on an adaptive group scoring mechanism integrated with blockchain technology to maintain a decentralized, secure, and immutable ledger of all transactions and data exchanges within the CAVs network. The blockchain ledger at this layer records all CAVs-related transactions and data exchanges in blocks secured by Kyber PQC to provide security against interception and tampering by quantum-capable adversaries. These blocks are structured in a chain, where each block contains three main elements: $\Sigma_{\text{Block}} = (\text{B_ID}, \text{Pr_B_Hash}, T_{\text{Block}})$, where B_ID serves as a unique identifier for the block. The element Pr_B_Hash represents the previous block's hash, establishing a cryptographic link that ensures continuity and security. T_{Block} comprises the set of validated transactions, which, in the case of the CAVs environment, might include data related to the vehicle status, authentication logs, or other critical transactional data such as traffic information. Thus, the consensus layer ensures robust and immutable data integrity, which is crucial for reliable and safe vehicular operations within the CAVs environment.

4. Cloud layer: This layer serves as a central repository within the CAVs environment that stores and processes a large volume of data remotely. It provides essential computational power and large-scale storage to support the complex data processing needs of CAVs. The cloud server handles non-critical data such as CAVs ownership details, cost, and origin information that does not compromise human safety if delayed. Once the edge server processes the critical data for immediate decision-making, it is transferred to the cloud for permanent storage, supporting future retrieval and long-term strategic planning and data analytics for CAVs. High-definition (HD) maps created using LiDARs and other sensors are stored here, including detailed environmental features such as lanes, road markers, and obstructions, which can assist in route planning and environmental perception, which can enhance navigation precision and safety.

4.2. Threat model and security measures

Given the exposure of CAVs to the Internet, they can be vulnerable to a variety of security threats where the adversaries could undermine the credibility of data shared among CAVs. The proposed system framework is designed to mitigate a wide array of attack vectors to ensure a resilient and secure CAVs network through several key security measures, which are described as follows:

- 1. Transactional integrity:** The framework ensures security against tampering and modifications to all network data and transactions by leveraging the cryptographic strength of Kyber PQC and the immutable nature of blockchain.
- 2. Identity anonymity:** Advanced encryption and anonymization methods are utilized to protect the identities of involved CAVs, ensuring that sensitive information remains confidential even during interception.
- 3. Data confidentiality:** The sensitive information is encrypted using Kyber PQC, which provides quantum-resistant encryption to secure the network against quantum and conventional cryptographic attacks.
- 4. Mutual authentication:** The consensus layer in our framework authenticates each transaction and communication within the network, thus verifying the legitimacy of each participating node and preventing unauthorized interventions.
- 5. Quantum-resistant security:** The proposed framework integrates Kyber PQC, designed to resist quantum computing attacks, including those facilitated by Shor's algorithm. This secures the network against sophisticated cryptographic attacks that could undermine existing encryption methods.

5. Proposed kyber-PQC-based-AGS-PBFT secure communication mechanism

5.1. Preparation phase

In the preparation phase, the Trusted Authority (TA) initializes the system by establishing a set of cryptographic primitives and parameters essential for the subsequent phases. The following system parameters are generated during this phase:

- The TA selects the polynomial ring $R_q = \mathbb{Z}_q[X]/(X^n + 1)$, where n , an integer indicating the degree of the polynomial and q is a large odd prime number, enabling efficient Number-Theoretic Transform (NTT) operations essential for rapid cryptographic computations.

Table 3
Cryptography symbols and descriptions.

Symbol	Description
pk	Public key
sk	Private/secret key
c	Cipher text
q	A sufficiently large prime number
n	An integer parameter in cryptographic functions
A	A public matrix used in the computation of public keys
H	Hash function
$ID_{CAV_{s_i}}$	Identity of i^{th} CAVs
ID_{rsu_i}	Identity of i^{th} RSU
MK	Master key of TA
B_n	Centered binomial distribution used for sampling keys
$Sign_c$	Signature of the digital proofs from client node
$Sign_l$	Signature of the digital proofs from leader node
$Sign_r$	Signature of the digital proofs from replica node
KDF	Key Derivation Function, specifically SHAKE-256 used for generating cryptographic keys
B_n	Noise parameter in key generation for security against lattice attacks
S_i	Security token computed using hash functions for authentication purposes

- The TA generates a private key (sk) by sampling from a noise distribution characterized by a centered binomial distribution B_n , ensuring resistance against lattice-based attacks. The private key components s and e are derived from this distribution as per Kyber specification. The public key, $pk := \hat{A} \cdot s + e$ is derived from the NTT-transformed private key components to facilitate secure message encryption.
- TA select a cryptographic hash function $H : \{0,1\}^* \rightarrow \{0,1\}^l$, where l is a fixed length output. The hash function is used for its robust pseudorandom properties and resistance to collisions.
- SHAKE-256 is employed as the Key Derivation Function (KDF) to securely generate multiple cryptographic keys from a single master key to provide necessary cryptographic diversity within the CAVs framework.
- The TA uses the Practical Byzantine Fault Tolerance (PBFT) algorithm employed within the blockchain consensus process. The parameters $\{n, q, B_n, H\}$ are published publicly to ensure transparency and integrity in the consensus process. The key notations used in this study are depicted in Table 3.

5.2. Registration phase

In the registration phase, the TA registers CAVs, RSUs, MEC servers, and cloud servers by utilizing the Kyber key encapsulation mechanism to ensure quantum resistance. The registration for each entity is described as follows:

5.2.1. CAVs registration

- The TA assigns a unique identifier $ID_{CAV_{s_i}}$ to each vehicle CAV_{s_i} .
- The TA generates a private key $sk_{CAV_{s_i}}$ and a corresponding public key $pk_{CAV_{s_i}}$ for the CAVs. This involves sampling secret key and error vectors $s_{CAV_{s_i}}$ and $e_{CAV_{s_i}}$ from a centered binomial distribution B_n , performing NTT on these vectors, and computing the public key as $pk_{CAV_{s_i}} = \hat{A} \cdot s_{CAV_{s_i}} + e_{CAV_{s_i}}$, where A is a public matrix generated from the system parameters.
- The TA computes a registration token $S_i = H(ID_{CAV_{s_i}}, pk_{CAV_{s_i}}, MK, TS_{CAV_{s_i}})$, where MK is the master key of the TA and $TS_{CAV_{s_i}}$ represents the current timestamp. This token S_i alongside the public key pk is securely stored within the TA's database and broadcasted to relevant network entities to facilitate secure communications. The private key sk remains securely stored in OBU of CAV_{s_i} , ensuring that it is protected from unauthorized access.
- The TA ensures that all transmitted data during registration, especially the public key and registration token are encrypted and

transmitted over secure channels to prevent interception and ensure data integrity.

5.2.2. RSU registration

- The TA assigns a unique identifier ID_{rsu_i} to each rsu_i . It then generates a private key sk_{rsu_i} and an error vector e_{rsu_i} from the centered binomial distribution B_n . The public key pk_{rsu_i} is computed using $pk_{rsu_i} = \hat{A} \cdot s_{rsu_i} + e_{rsu_i}$ for each RSU.
- To facilitate secure communication and authentication, the TA computes a registration token $S_{rsu_i} = H(ID_{rsu_i}, pk_{rsu_i}, MK, TS_{rsu_i})$, where MK is the master key of the TA and TS_{rsu_i} is the timestamp marking the registration. The TA transmits the message containing $(ID_{rsu_i}, S_{rsu_i}, pk_{rsu_i})$ to rsu_i through a secure channel.
- Upon receiving the message from the TA, rsu_i generates its own private key sk_{rsu_i} from the same distribution B_n and computes its corresponding public key pk_{rsu_i} . It stores the registration token S_{rsu_i} , its private key sk_{rsu_i} , and other relevant information securely in its database. The rsu_i publishes its public key pk_{rsu_i} publicly to enable secure communications with other network entities. The private key sk_{rsu_i} is securely stored within the rsu_i , ensuring its confidentiality.

5.2.3. MEC server registration

- For each mobile edge computing server mec_i , TA generates a private key sk_{mec_i} and public key $pk_{mec_i} = \hat{A} \cdot sk_{mec_i} + e_{mec_i}$. For each mec_i , TA computes a registration token $S_{mec_i} = H(ID_{mec_i}, pk_{mec_i}, MK, TS_{mec_i})$, incorporating the master key MK and a timestamp TS_{mec_i} . It transmits this to mec_i via a secure communication channel.
- Upon receiving the registration details from TA, mec_i generates its private key sk_{mec_i} using the same distribution B_n and computes its corresponding public key pk_{mec_i} . This token and the private key are securely stored within its database and it publishes the pk_{mec_i} to relevant entities, facilitating secure communications.

5.2.4. Cloud server registration

- For each cloud server cs_i , TA generates a secret key sk_{cs_i} and computes the public key $pk_{cs_i} = \hat{A} \cdot sk_{cs_i} + e_{cs_i}$. A registration token $S_{cs_i} = H(ID_{cs_i}, pk_{cs_i}, MK, TS_{cs_i})$ is calculated, using the master key MK and a timestamp TS_{cs_i} . The registration details are sent to cs_i through a private channel. Upon receiving the registration details, cs_i generates its private key and computes its public key. The token and secret key sk_{cs_i} are securely stored in the cloud server, and cs_i publishes its pk_{cs_i} with other network entities.

5.3. Authentication and session key establishment for V2R and V2V communications

This phase utilizes a Kyber cryptographic protocol to authenticate the Vehicle-to-Roadside unit (RSU) and Vehicle-to-Vehicle (V2V) communication. It ensures secure communication where the entities within the network verify each other's identities before exchanging sensitive information and establishing shared session keys, as described in Algorithm 1.

5.3.1. V2R communication

V2R communication ensures that CAVs can efficiently exchange information with roadside units (RSUs) to enhance road safety and traffic efficiency. The following describes how each vehicle CAV_{s_i} and RSU rsu_i employs the Kyber key generation process to secure the communication:

- Each vehicle CAV_{s_i} or RSU rsu_i generates its cryptographic keys using the Kyber key generation process which involves generating a seed d and deriving parameters (ρ, σ) using the generator function $G(d)$. It produces a public matrix $\hat{A}$ in the Number Theoretic

Algorithm 1 Secure authentication and key establishment using Kyber.

```

1: Input:  $ID_{\text{entity}}$ : Unique identifier for each entity
2: Input: System-wide cryptographic parameters:
3:  $n$ : Polynomial degree
4:  $q$ : Modulus for coefficients
5:  $\eta_1, \eta_2$ : Noise distribution parameters
6: Output: Shared secret session key  $K$ 
7: procedure KEYGENERATION
8:   Generate seed  $d$  and derive  $(\rho, \sigma) \leftarrow G(d)$ 
9:   for each  $i$  from 0 to  $k-1$  do
10:     Generate matrix  $\hat{A}[i] \leftarrow \text{Parse}(XOF(\rho, i))$  in NTT domain
11:     Sample  $s[i]$  using  $B_{\eta_1}$  and  $e[i]$  using  $B_{\eta_2}$ 
12:      $s \leftarrow \text{NTT}(s)$  and  $e \leftarrow \text{NTT}(e)$ 
13:   end for
14:    $pk \leftarrow \hat{A} \cdot s + e$ 
15:    $sk \leftarrow (s, e, pk, \text{Hash}(pk), \sigma)$ 
16: end procedure
17: procedure PUBLICKEYEXCHANGE
18:   Compute token  $S_i \leftarrow H(ID_{\text{entity}}, pk_i, MK, TS)$ 
19:   Exchange  $pk$  and  $S_i$  securely
20: end procedure
21: procedure AUTHENTICATIONKEYEXCHANGE
22:   On receiving  $pk_j$  and  $S_j$ , verify  $S_j$ 
23:   Abort if verification fails
24:   Generate random  $m, r$ ; compute  $m \leftarrow H(m)$ 
25:   Encapsulate:  $(c, k_r) \leftarrow (\text{NTT}^{-1}(\hat{A}^T \cdot r + e), H(k_i \parallel H(c)))$ 
26:   Send  $c$ ; On receiving  $c$ , decapsulate to compute:
27:    $m' \leftarrow \text{NTT}^{-1}(\text{NTT}(s)^T \cdot \text{NTT}(r))$ 
28:    $k'_r \leftarrow H(k_i \parallel H(c))$ 
29:   Verify  $k_r$  matches  $k'_r$ ; initiate error protocol if mismatch
30: end procedure
31: procedure ESTABLISHSESSIONKEY
32:   Both parties compute  $K \leftarrow H(k_i \parallel k'_j)$ 
33: end procedure

```

Transform (NTT) domain from the seed ρ using the Extendable Output Function (XOF) for efficient polynomial arithmetic. The secret vector $s[i]$ and noise vector $e[i]$ are sampled from the binomial distribution B_{η_1} , and transformed into the NTT domain.

- The public key is computed as, $pk = \hat{A} \cdot s + e$ which is broadcasted securely within the network. The secret key is computed as $sk = (s, e, pk, \text{Hash}(pk), \sigma)$ which is stored. A security token S_i is computed as a hash for each entity as $S_i = H(ID_{\text{entity}}, pk_i, MK, TS)$, facilitating authenticated communications, safeguarding against replay attacks. The public key pk and its associated token S_i are exchanged securely among entities.
- Upon receiving a public key $pk_{CAV s_i}$ and a security token S_i from a vehicle $CAV s_i$, the Roadside Unit rsu_i verifies the security token to ensure the public key's authenticity. If the verification is successful, rsu_i generates a random message m and a random value r , and computes $m = H(m)$, where H is a cryptographic hash function.
- An encapsulation operation is performed by rsu_i , where a ciphertext c and a temporary key k_{rsu_i} are generated. This is done using the inverse Number Theoretic Transform (NTT) of the product of the transposed matrix $\hat{A}^T$ and r plus e , encrypting the message under the vehicle's public key $pk_{CAV s_i}$.
- The ciphertext c is sent to $CAV s_i$, who then decapsulates it to compute: $m' = \text{NTT}^{-1}(\text{NTT}(s_{rsu_i})^T \cdot \text{NTT}(r))$ and to derive $k'_{CAV s_i} = H(k_{rsu_i} \parallel H(c))$ where $\parallel$ denotes the concatenation operation.
- The integrity of the operation is confirmed by ensuring that k_{rsu_i} matches $k'_{CAV s_i}$, which would indicate successful authentication and key agreement.

5.3.2. Secure session establishment

- A session key is established using the shared secret K through a Key Derivation Function (KDF), facilitating secure communication.

5.3.3. Vehicle-to-vehicle communication

- If vehicles $CAV s_i$ and $CAV s_j$ communicate through their respective RSUs rsu_i and rsu_j . The shared keys between $CAV s_i, rsu_i$ and $CAV s_j, rsu_j$ are established using the RSUs' public keys. rsu_i encapsulates $K_{rsu_i, CAV s_i}$ for rsu_j , which on decapsulation, re-

encapsulates it for $CAV s_j$. Upon decapsulation, $CAV s_j$ now shares a key initially established between $CAV s_i$ and rsu_i .

5.4. Sharing and storing of traffic information using AGS-PBFT

This phase leverages the Adaptive Grouping Score-based Practical Byzantine Fault Tolerance (AGS-PBFT) mechanism for quantum-resistant operations in the CAVs blockchain network. The principal role of AGS-PBFT is to enhance the reliability of consensus nodes by dynamically adjusting group sizes based on node performance metrics. It selects an entity to participate in the consensus mechanism and store their information in the blockchain network to ensure secure, transparent, and immutable recording of traffic information, telemetry data, and other critical data exchanges between CAVs and RSUs, thereby enhancing overall traffic management and vehicle coordination.

The process initiates when a CAV or RSU, acting as a client, generates a transaction request, including the transaction ID, entity ID, and service request details. This request is then processed through the following AGS-PBFT consensus mechanism stages, ensuring that multiple nodes verify and agree upon all operations to ensure integrity and security against malicious alterations. The AGS-PBFT process for our proposed framework is summarized in Algorithm 2, where the system initializes the blockchain ledger and generates and distributes cryptographic keys based on the Kyber PQC to all nodes, enabling secure communication and verification processes across the network. The AGS-PBFT process involves the following stages:

- Initialization:** The nodes are initialized with a score of 100 and divided into a group of consensus and candidate node sets randomly. The consensus nodes will execute the consensus process and the candidate nodes will only update their local states during consensus. Half of the total nodes, $N/2$ are candidate nodes, and the number of consensus nodes CN , satisfies $CN \geq 3f + 1$, where f is the maximum number of Byzantine nodes.
- Dynamic Grouping:** Nodes are dynamically grouped based on their scores and network conditions. Smaller groups are formed during low activity to reduce overhead, while larger groups are formed during high activity for better fault tolerance. For instance, with 100 nodes, we might have two groups of 25 nodes each during low activity, while during high activity, a single group of 50 nodes might be used. The dynamic adjustment of group sizes improves computational efficiency and ensures system robustness under varying network traffic.
- Leader Node Selection:** The leader node, also known as the primary node, is selected using $p = v \bmod CN$. If the leader node fails, then the view will be converted, and the next leader node will be selected for replacement.
- Request:** The client node initiates a transaction request, identified by ID_i , to the leader node.
- Pre-Prepare:** The leader node constructs a pre-prepare message and encodes the request message, including the request, the view number (v), and sequence number (s_n), along with the leader node signature ($Sign_l$) and broadcasts to all replica nodes.
- Prepare:** Upon receiving the pre-prepare message, each replica node verifies the information using Kyber PQC to ensure the request authenticity, and if the verification is successful, it broadcasts the prepare message, including their signature ($Sign_r$) to consensus nodes. If verification passes, nodes will enter the response stage.
- Response:** The replica node enters the response stage, verifies the message and returns the result to the leader node.
- Reply:** When the leader node receives more than $(2f+1)$ identical confirmation messages, it is considered that a consensus has been reached. The leader node then provides the confirmation results to all nodes, including candidate nodes, and updates the scores of all nodes.

Algorithm 2 Enhanced Quantum-resistant AGS-PBFT for CAVs.

```

1: Inputs:
2:  $ID_i$  - Unique identifier for each CAVs node
3:  $REQ$  - Request from the client (CAVs or RSU)
4:  $KYBER-PQC$  - Cryptographic scheme for digital signatures and verification
5: Outputs:
6: Confirmation and logging of transaction in the blockchain ledger
7: procedure INITIALIZATION
8:   Initialize blockchain ledgers on all nodes
9:   Generate and distribute  $KYBER-PQC$  cryptographic keys
10:  Elect a leader using a leader-selection method specific to the network dynamics
11:  Initialize all nodes with a score of 100 and divide into consensus and candidate
    sets randomly
12: end procedure
13: procedure REQUEST( $ID_i, REQ$ )
14:   Client vehicle signs  $REQ$  with its private key using  $KYBER-PQC$  to create  $\delta_{client}$ 
15:   Client vehicle broadcasts  $\langle REQ, Sign_{\delta} \rangle$  to the primary node (leader) using  $ID_i$ 
16: end procedure
17: procedure PRE-PREPARE( $REQ, v, s_n$ )
18:   Primary node (leader) signs  $REQ$  using  $KYBER-PQC$ 
19:   Broadcasts  $\langle PRE-PREPARE, v, s_n, REQ, Sign_{\delta} \rangle$  to all replica nodes
20: end procedure
21: procedure PREPARE( $\langle PRE-PREPARE, v, s_n, REQ, Sign_{\delta} \rangle$ )
22:   Replicas verify the  $PRE-PREPARE$  message using  $KYBER-PQC$ 
23:   Upon successful verification, each replica broadcasts  $\langle PREPARE, v, s_n, Sign_{\delta} \rangle$ 
24: end procedure
25: procedure RESPONSE( $\langle PREPARE, v, s_n, Sign_{\delta} \rangle$ )
26:   Verify all  $PREPARE$  messages from at least  $2f + 1$  replicas
27:   On obtaining sufficient valid  $PREPARE$  messages, broadcast
     $\langle RESPONSE, v, s_n, Sign_{\delta} \rangle$ 
28: end procedure
29: procedure REPLY
30:   Execute the transaction upon receiving  $2f + 1$   $RESPONSE$  messages
31:   Update blockchain: append the new block containing transaction details
32:   New block B:  $\sum_{Block} = (B\_ID, Pr\_B\_Hash, T_{Block})$ 
33:   B_ID - Unique block identifier, Pr_B_Hash - Hash of the previous block
34:    $T_{Block}$  - Transactions validated in this block
35:   Notify the client of the transaction result securely using  $KYBER-PQC$ 
36: end procedure
37: procedure SCORE UPDATE AND GROUP ADJUSTMENT
38:   if  $\exists \sum(\{result[i] \in result\}) > (2f + 1)$  then
39:     For nodes with consistent confirmation results, increase scores by 1
40:     For nodes with inconsistent results, reduce scores by 5
41:     The primary node broadcasts the result{i}, and the consensus nodes will update
    their local state
42:   else
43:     Do nothing
44:   end if
45:   count_request++
46:   if count_request == 50 then  $\triangleright$  Periodic node reassignment and adjustment
47:     Move the lowest-scoring nodes from the consensus set to the candidate
48:     Move the highest-scoring nodes from the candidate set to the consensus
49:     count_request = 0  $\triangleright$  Reset request counter after updating nodes.
50:   end if
51:   Adjust group sizes based on network conditions and node performance metrics
52:   Use smaller groups during low network activity to reduce overhead, else use larger
    groups during high network activity for better fault tolerance
53: end procedure

```

- **Score Update and Group Adjustment:** The AGS-PBFT mechanism dynamically adjusts node scores and group sizes to enhance network efficiency and fault tolerance. Each node is initialized with a score of 100, and the node scores are then updated based on their agreement with the consensus result. For nodes that agree with the consensus result, scores are raised by one, and for nodes that disagree with the result, scores are lowered by five. This scoring process incentivizes correct behavior and penalizes malicious behavior.

To ensure dynamic adaptation, the thresholds are determined based on the mean (μ) and standard deviation (σ) of the scores at the end of each reassignment, where μ represents the average score of all nodes, and σ denotes the measure of variation in scores across the nodes. Nodes with scores below a lower threshold ($T_{low} = \mu - \sigma$) are allocated to the candidate node set, while nodes with scores above a higher threshold ($T_{high} = \mu + \sigma$) are assigned to the consensus node-set. Nodes are reassigned between consensus and candidate sets based on updated scores periodically, i.e. every 50 requests. Furthermore, to optimize resource utilization and ensure system robustness, the size of the consensus group is dynamically adjusted

based on network activity, which reduces computational overhead and enhances fault tolerance and system stability.

The transaction is then executed, and the new data block containing the validated transaction is added to the blockchain as $Block = (B_ID, Pr_B_Hash, T_{Block})$. This block is then linked cryptographically to previous blocks, thereby ensuring the integrity and continuity of the blockchain. The client is notified of the transaction outcome, and Kyber PQC is utilized for secure communication.

Thus, the AGS-PBFT protocol provides a robust framework for high-throughput, low-latency transaction processing in vehicular networks, ensuring that all recorded transactions are verifiable and auditable, fostering trust among network participants.

6. Security analysis of the proposed approach

This section presents the analysis of the proposed approach for facilitating secure communication in a CAVs environment. The proposed approach is protected against various quantum and existing attacks, ensuring robust and reliable security features for CAVs. In the following, we describe the key security features.

1. **Forward secrecy:** The key generation procedure of Kyber is efficient, allowing for frequent key renewals. This ensures that previously encrypted communications remain secure even if a private key is compromised. Thus, during each key generation, the seed d is used to derive two parameters ρ and σ using a pseudorandom generator function, $(\rho, \sigma) \leftarrow G(d)$. Using ρ , a unique matrix $\hat{A}$ is generated through an extendable output function (XOF) defined as $\hat{A} = \text{Parse}(XOF(\rho, i))$, $i \in \{0, \dots, k-1\}$. Further, the public key is computed as, $pk = \hat{A} \cdot s + e$ where s and e are secret and error vectors sampled from the centered binomial distribution B_{η} . Since ρ is unique for every session, each public key is independent, thereby ensuring forward secrecy.

2. **Anonymity:** It refers to the capability to hide the true identities of entities (such as CAVs or infrastructure units) participating in communication sessions. This is essential for maintaining privacy and preventing unauthorized entities from identifying during communications. To ensure anonymity, our proposed approach uses encrypted identifiers rather than actual identities, which protects the real identities of entities during communication sessions. Thus, for each CAV, the identity ID_{CAV_i} is combined with other parameters to generate a secure token: $S_i = H(ID_{CAV_i} \parallel pk_{CAV_i} \parallel MK \parallel TS)$ where $H(\cdot)$ is a cryptographic hash function, MK is the master key, and TS is the current timestamp. The use of a one-way hash function $H(\cdot)$ ensures that even if an attacker obtains S_i , it is computationally infeasible to recover the original identity ID_{CAV_i} .

3. **Primal attacks:** These attacks involve constructing a unique Shortest Vector Problem (SVP) instance from the Learning With Errors (LWE) problem and solving it using the Blockwise Korkin-Zolotarev (BKZ) algorithm. Thus, Kyber uses Module-LWE instead of Ring-LWE, making it more resistant to primal attacks. The security parameters ensure that the block dimension required for a successful BKZ attack is computationally infeasible.

4. **Dual attacks:** These attacks seek to find a short vector in the dual lattice to exploit the correlation between errors and secrets. In Kyber, the secret key s and the error term e are sampled from a centered binomial distribution B_{η} , ensuring that both are random and independent. Since the dual attack attempts to find short vectors in the dual lattice of $\hat{A}$, it is computationally hard due to the randomness of s and e . Moreover, the parameters of the Module-LWE problem are chosen such that the dimension and modulus q increase the security of the system. Thus, the parameters in Kyber's Module-LWE design ensure that finding such a short vector is highly unlikely.

5. **Multi-target attacks:** This attack aims to compromise one out of many possible targets by attacking multiple keys simultaneously. Kyber mitigates these attacks through matrix regeneration, i.e. for each public key, the matrix $\hat{A}$ is regenerated, ensuring that multi-target at-

Table 4
Simulation parameters.

Parameter	Description
Network simulation tool	OMNET++ 6.0
Traffic simulation tool	SUMO 1.20.0
Mobility generation tool	Veins 5.2
Cryptographic library	PQClean
Blockchain framework	Hyperledger Fabric 2.2
Simulation area	5 km × 5 km
Simulation time	600 s
Speed of vehicle	10 m/s (36 km/h)
Vehicle density	50-300 vehicles
Data transmission rate	6 Mbps
Transmission power	20mW
NoiseFloor	-98dBm
Consensus algorithm	AGS-PBFT
No. of consensus nodes	10-15
Post-quantum cryptography mechanism	Kyber
Hashing algorithm	SHAKE-256
Wireless protocol	IEEE 802.11p
Network topology	Urban grid

tacks are infeasible. The generation of the public matrix $\hat{A}$ is achieved as, $\hat{A} = \text{Parse}(XOF(\rho, i))$ where, ρ is derived from a unique seed d for each session, ensuring that the matrix $\hat{A}$ is distinct for every public key. This protects against an attacker attempting to break many keys by breaking one key.

6. Misuse resilience: It refers to the system's ability to remain secure even when misused. Kyber operates under IND-CCA2 security, ensuring that the system remains secure even if an attacker can choose ciphertexts to decrypt. IND-CCA2 security allows Kyber to be safely used with static keys and to reuse ephemeral keys without compromising security. The CCA transform protects against misuse by hashing the public key and ciphertext into the final key, ensuring contributory behavior and robustness.

7. Timing attacks: The Kyber implementations avoid secret-dependent branches and table lookups, which are common sources of timing leakage. This is achieved through constant-time instructions and careful handling of multiplications and modular reductions. Operations like NTT(s) and modular reductions [25] are implemented using techniques like Montgomery and Barrett reductions to avoid timing leaks. The NTT transformation converts polynomials into the frequency domain as, $\text{NTT}(f(X)) = \sum_{i=0}^{n-1} f(X) \cdot \omega^{ij} \mod q$ where ω is the primitive n -th root of unity modulo q . Thus, operations in the NTT domain are faster and more secure than direct polynomial multiplications.

7. Performance evaluation and discussions

The performance of the proposed Kyber-based-AGS-PBFT approach is assessed using a vehicular networking environment. For the simulation experiment, we utilized the traffic flow simulation tool SUMO 1.20.0, Veins 5.2 for vehicular mobility simulations, and OMNET++ 6.0 for networking. Additionally, we have used OpenStreetMap (OSM) [50] to generate a realistic road network for the simulation, and the simulation area covering a 5 km × 5 km grid is selected, which includes intersections, road segments and traffic flows modeled after real-world urban environments of Chandigarh city [51]. To simulate the vehicle movement, we have employed the RandomTrips Python script to generate random vehicle trips. The SUMO configuration file was edited to integrate the network, topographic, and vehicle trip files. This configuration allows us to evaluate the impact of mobility patterns on the scalability and efficiency of the proposed approach. The influence of network density is analyzed by varying the number of vehicles from 50 to 300 and observing its impact on end-to-end latency, throughput, and communication overhead. Further, to ensure quantum-resistant secure communications, we have integrated the PQClean library into our OMNET++ simulation to implement post-quantum cryptographic operations. It provided strong encryption with minimal computational

Table 5
Parameters for Kyber.

Parameter	n	k	q	η_1	η_2	sk (B)	pk (B)	c (B)
Value	256	4	3329	2	2	3168	1568	1568

overhead, making it a scalable solution for secure CAV environments. The parameters used in the simulations are shown in Table 4 and 5. The SUMO simulation from the real map and Veins simulation are shown in Fig. 4.

7.1. Computation cost analysis

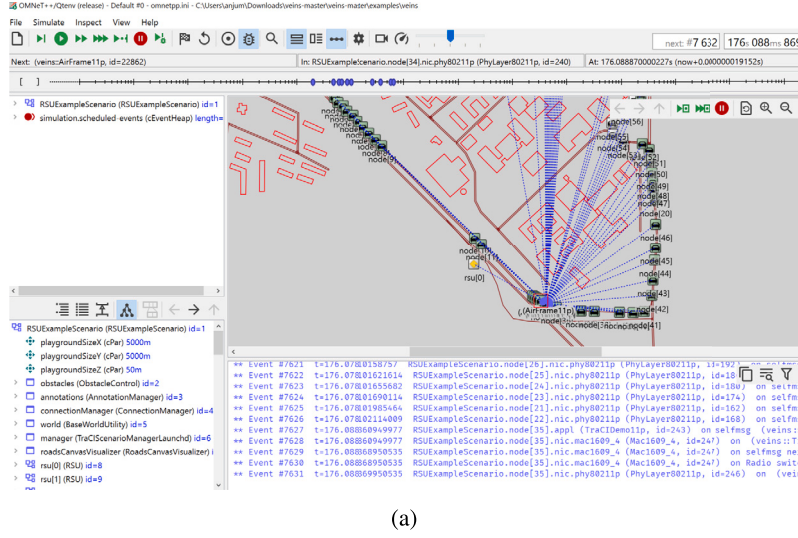
The performance of the proposed Kyber-based-AGS-PBFT approach is evaluated and compared with the existing and other post-quantum solutions, which include [28], [29], [30], [31]. Table 6 compares the proposed approach with various methods in terms of computation overhead, which is visually illustrated in Fig. 5. As shown in the Table 6, $T_{\text{ecm}}, T_{\text{epa}}, T_h, T_{\text{ecc}}, T_{\text{ecc}}^d, T_{\text{e-d}}, T_{\text{vm}}, T_{\text{vv}}, T_{\text{sm}}, T_{\text{va}}, T_{\text{vv}}, T_{\text{mm}}, T_e, T_d, T_B$ denotes the time needed for “elliptic curve point multiplication”, “elliptic curve point addition”, “hash function”, “elliptic curve point encryption”, “elliptic curve point decryption”, “key encryption/decryption”, “vector-to-matrix multiplication”, “vector-to-vector multiplication”, “scalar-vector multiplication”, “vector addition”, “vector-to-vector multiplication”, “matrix-to-matrix multiplication”, “symmetric encryption”, “symmetric decryption”, “read-time from blockchain”.

The total computations of the proposed approach are computed as $2T_{\text{mm}} + 4T_{\text{vm}} + 6T_{\text{vv}} + 6T_h + T_e + T_d + 2T_B$, where matrix-to-matrix multiplications are required for the key generation and encryption processes in Kyber. Vector-to-matrix and vector-to-vector multiplications are used during the encryption and decryption process, where the secret vector is multiplied by the public matrix to generate the cipher text. Hash functions are used to generate keys and ensure integrity to provide security. Further, in the blockchain consensus mechanism, the transaction undergoes several validation and consensus steps. The coefficient 2 with T_B reflects the dual operations required for initial transaction verification and another for achieving consensus on the transaction within the blockchain. Furthermore Table 7 provides a comparative analysis of the computational requirements of Kyber PQC with classical cryptographic algorithms, highlighting significant improvements particularly in terms of computational complexity and key generation efficiency. Thus, the proposed solution demonstrates lower computation overhead than [28], [29], [30], [31] approaches, making it a viable solution for resource-constrained CAVs.

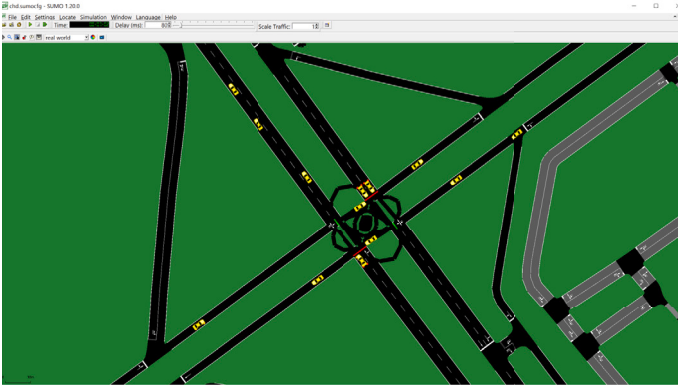
7.2. End-to-end latency

The end-to-end latency is defined as the average time taken for messages to travel from source to its destination across a network. This metric can be expressed as: $\sum_{i=1}^M (T_{rv_i} - T_{sd_i}) / M$, where M is the total number of messages and T_{rv_i} and T_{sd_i} represent the received and sent times of packet i . In our proposed approach, the end-to-end latency includes the time required for authentication and session key establishment between communicating entities. We analyze the impact of network density by varying the number of vehicles from 50 to 300. Fig. 6a illustrates the latency for the proposed approach compared with [28], [29], [30], and [31] approaches. As depicted, the communication delay in our approach is about 920.70 ms for 50 vehicles and rises to 1751.20 ms when the number of vehicles reaches 150 and approximately 2300 ms for 300 vehicles. In contrast, other approaches exhibit higher latencies of 2997.60 ms, 4310.80 ms, 5200 ms, and 5980.50 ms, respectively.

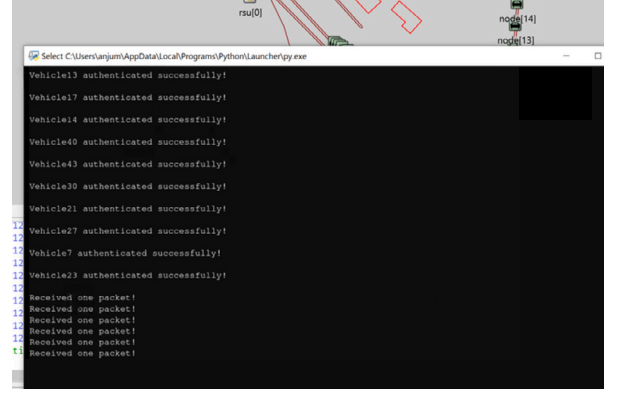
It is evident that the communication delay for all approaches increases almost linearly with the increase in network density. However, the proposed approach facilitates efficient message propagation due



(a)



(b)



(c)

Fig. 4. SUMO and Veins simulation for Kyber-based AGS-PBFT approach: (a) SUMO-based traffic flow simulation of CAVs, (b) OMNET++ simulation displaying the communication between V2V and V2I, and (c) Vehicle authentication process.

Table 6

Comparative computation cost analysis.

Approaches	Cryptosystem	Total computations	Computation cost (ms)
[31]	Elliptic curve	$13T_{ecm} + 3T_{epa} + (8 + 2i)T_h + 2T_{ecc}^e + 2T_{ecc}^d + 5T_{e-d}$	≈ 359.86
[30]	Elliptic curve	$10T_h + 9T_{ecm} + 4T_{epa}$	≈ 234.75
[29]	Lattice	$5T_{vm} + 2T_{vv} + 6T_{sm} + 4T_h + 4T_{va}$	≈ 16.67
[28]	Lattice	$4T_{vm} + 2T_{vv} + 6T_{sm} + 4T_h + 4T_{va}$	≈ 14.64
Ours	Lattice	$2T_{mm} + 4T_{vm} + 6T_{vv} + 6T_h + T_e + T_d + 2T_B$	≈ 10.31

to the adaptive nature of the Kyber-based AGS-PBFT, which ensures a significant performance improvement, maintaining scalability and efficiency even in highly dynamic and high-density vehicular environments compared to other approaches.

7.3. Impact on throughput

Throughput is defined as the number of bits transmitted per unit of time. Fig. 6b illustrates the throughput of the proposed approach compared with other approaches. It can be computed as $p_{rv} \times |p|/T$, where p_{rv} is the total number of received packets, T is the total time (in seconds), and $|p|$ is the size of a packet. For our simulation, the total time is 600 seconds. It is evident that as the number of vehicles increases, the throughput also increases due to the larger number of messages being exchanged. The proposed approach handles the increased communica-

tion effectively and increases overall computing throughput by 59.30% compared to [28], 71.32% compared to [29], 76.47% compared to [30], and 78.68% compared to [31]. This enhancement is attributable to the efficient cryptographic and consensus mechanisms integrated within the Kyber-AGS-PBFT system, which optimize data flow and processing speed even as the network scales. Furthermore, the impact of mobility patterns on throughput is assessed using the SUMO and Veins tools which exhibit robust throughput performance even under bursty and random movement patterns.

7.4. Communication round analysis and security attribute comparison

In the CAV environment, a large number of messages are exchanged between the different entities. As depicted in Fig. 6c, when the number of vehicles is increased to 300, our approach reduces the number

Table 7
Computational requirements of Kyber PQC compared to classical cryptographic algorithms.

Requirements	Kyber PQC	Classical cryptographic algorithms	Significance
Key generation	Number Theoretic Transform (NTT) for efficient polynomial multiplication, binomial distribution sampling.	RSA: Large prime generation, modular exponentiation [43], ECC: Elliptic curve point generation using scalar multiplication [44].	Kyber PQC provides faster key generation compared to classical cryptographic algorithms, efficient real-time key exchange in CAVs, and quantum-resistance.
Encryption process	NTT-based polynomial operations (vector-to-matrix multiplication), vector addition, noise addition for security, hash-based IND-CCA2 secure encryption.	RSA: Modular exponentiation, ECC: Elliptic curve point multiplication	Faster encryption process, ensures IND-CCA2 security to protect against adaptive ciphertext attacks.
Decryption process	Inverse NTT computation to convert polynomials, noise subtraction, polynomial modular arithmetic.	RSA: Modular exponentiation, ECC: Elliptic curve point multiplication.	Kyber PQC has reduced computational complexity to $O(n \log n)$, ensuring faster decryption, suitable for resource-constrained CAVs application.
Computational complexity	$O(n \log n)$ due to efficient NTT-based polynomial operations.	RSA: $O(n^3)$ for modular exponentiation, ECC: $O(n^2)$ for elliptic curve scalar multiplication.	Kyber PQC is significantly faster than classical cryptographic algorithms, efficient scaling for larger key sizes.
Key size	Public: 1568 bytes, Private: 3168 bytes.	RSA: 2048–4096 bits (public key), ECC: 256–512 bits (public key).	Kyber PQC key size is larger than classical algorithms, supports resistance to quantum attacks.
Memory requirements	Memory required for Polynomial buffers for NTT operations and noise vectors.	Minimal buffer requirements for modular arithmetic.	Kyber requires more memory but enables faster computation, enhances security through polynomial-based randomness.
Noise distribution	Binomial noise sampling.	NA	Adds hardness to the MLWE problem, ensures resistance against quantum adversaries.

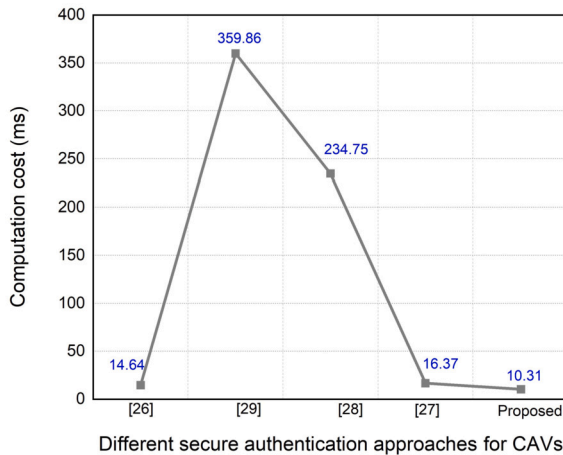


Fig. 5. Comparative analysis of computation cost wrt. different approaches.

of communication rounds compared to [28], [29], [30], and [31]. This substantial reduction in communication rounds directly contributes to minimized communication delays. As a result, our approach sustains high efficiency in completing authentication and key establishment processes, even amidst dense vehicular environments, thus ensuring robust and efficient CAVs communication.

Fig. 6d illustrates the performance comparisons of the proposed approach with the existing approaches in terms of security levels achieved. We have considered various security features as depicted in Table 2 in the comparative analysis. Thus, as per the analysis, the proposed approach offers enhanced security and resistance against quantum and existing attacks.

8. Conclusion, limitations, and future work

This study proposed a hybrid security solution integrating the Kyber Post-Quantum Cryptography with Adaptive Grouping Score-based Prac-

tical Byzantine Fault Tolerance (AGS-PBFT) for enhancing security in Connected and Autonomous Vehicles (CAVs). The comparative analysis has demonstrated substantial improvements, significantly reducing latency up to 47.16%, achieving an approximate 65.6% overall reduction in computation costs and effectively increasing the average throughput by 71.44% compared to existing security solutions. The integration of Kyber PQC ensures that our approach is designed to provide resistance to quantum computational attacks, providing a future-proof solution in anticipation of quantum computing advancements. Meanwhile, the AGS-PBFT mechanism adapts to dynamic vehicular network conditions, enhancing scalability and efficiency in consensus processes, which are crucial for maintaining robust and reliable vehicular communications. Although this study offers valuable insights, certain limitations should be acknowledged. First, the **computational overhead of post-quantum cryptography** may present challenges in real-time vehicular communication scenarios, where low latency is crucial. Additionally, the **energy demands of lattice-based schemes** could impact the feasibility of long-term deployment in resource-constrained CAV environments. Finally, **large-scale experimental validation is necessary to fully assess the scalability and performance of the proposed solution in real-world vehicular networks**. Future works can focus on expanding the cryptographic robustness by exploring other lattice-based cryptographic frameworks and integrating other primitives from NIST, such as Dilithium and Saber may offer different trade-offs between security and performance. Another research direction is to integrate machine learning and deep learning techniques to predict and adapt to dynamic network conditions in real-time.

CRedit authorship contribution statement

Anjum Mohd Aslam: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Aditya Bhardwaj:** Writing – review & editing, Visualization, Validation, Supervision, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Rajat Chaudhary:** Writing – review & editing, Validation, Supervision, Software, Conceptualization.

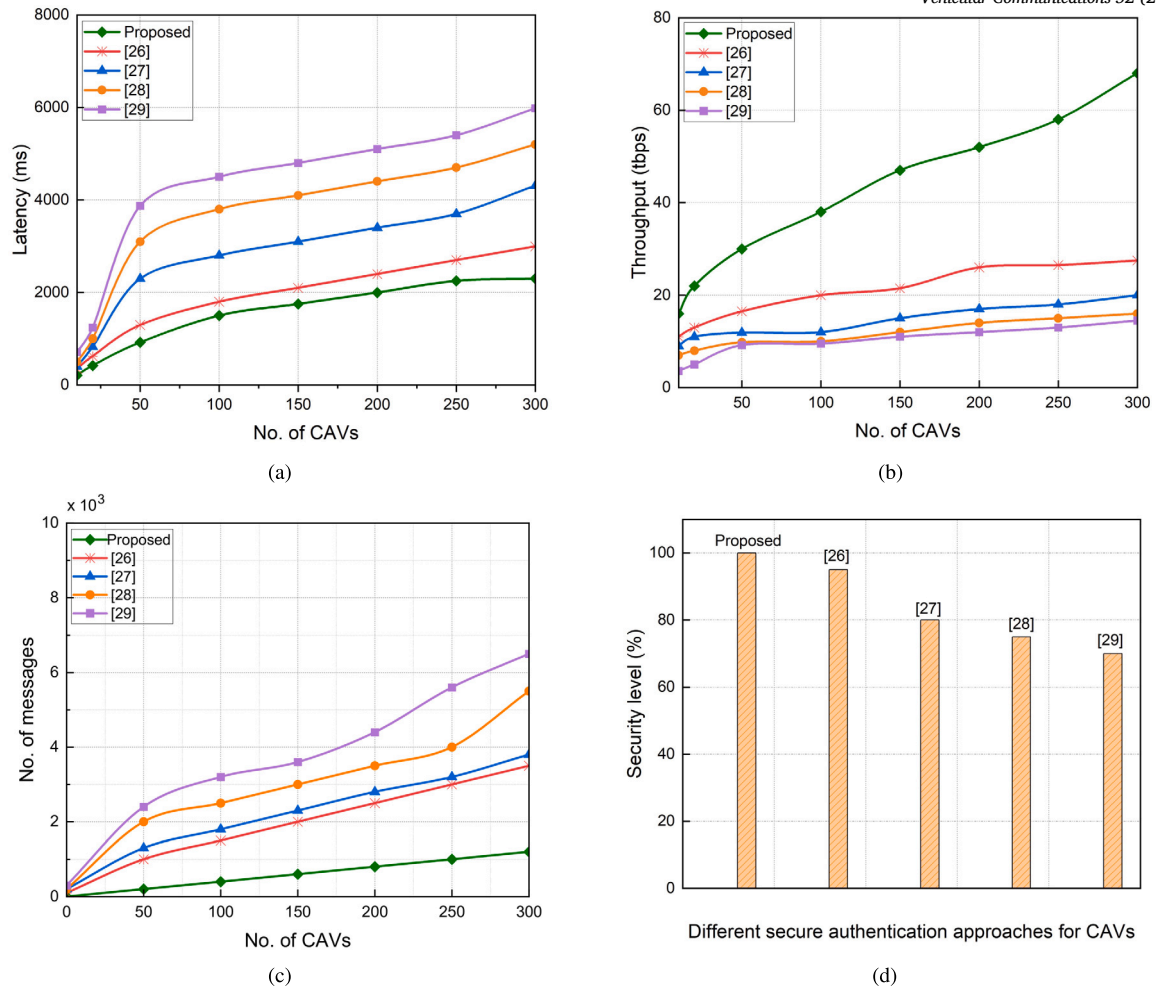


Fig. 6. Performance comparisons of the proposed approach with various existing authentication approaches for CAVs: (a) End-to-end latency vs. number of CAVs, (b) Throughput vs. number of CAVs, (c) Communication overhead vs. number of CAVs, and (d) Comparisons of security levels (%) wrt. different approaches.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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